

Electron EL-UHF-RMT01 Protocol Documentation

version

Electron Indonesia

January 21, 2021

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Welcome to Electron EL-UHF-RMT01's documentation!

Electron EL-UHF-RMT01 is a UHF RFID Reader with small dimension, low power consumption and 3.3V TTL UART Communication.

EL-UHF-RMT01 consists of an Microcontroller Unit and an antenna which acts as an integrated RFID Reader Writer, and is supposed to be controlled by a computer or another Microcontroller Unit. In this document, the computer or Microcontroller Unit will be called as **Host Computer** and the EL-UHF-RMT01 will be called as **Interrogator**.

The UART data format is 8 data bits, no parity, and 1 stop bit (8N1).

Serial communication between the **Host Computer** and the **Interrogator** will be documented in hexadecimal. All data in the frame tables are represented in hexadecimal.

Frame Definition

The serial communication between the **Host Computer** and the **Interrogator** is done frame by frame with formats explained below.

Frame Format

Header	Type	Instruction Code	Parameter Length (PL)	Parameter	CRC	End
BB	00	07	00 01	01	09	7E

The CRC is calculated by summing the frame bytes from Type to the last byte of Parameter, only taking the LSB byte.

Frame Type

Type	Description
0x00	Command from the Host Computer to the Interrogator
0x01	Response from the Interrogator to the Host Computer
0x02	Notification from the Interrogator to the Host Computer

Each **command frame** has a corresponding **response frame**. The **response frame** indicates whether the instruction has been executed.

There are corresponding **notification frames** for single inventory instructions and multiple inventory instructions. The **notification frames** is automatically sent to the **Host Computer** by the **Interrogator** according to the reading situation. When the **Interrogator** reads a tag, it sends a **notification frame**, and when the **Interrogator** reads multiple tags, it sends multiple **notification frames**.

Frame Code Index

This is the list of possible code for the command frame and the corresponding response frame. Usually, the response frame will have the same code as the command code with this exception:

- 0xE5 command code can have 0xE5 or 0xE6 response code
- 0xD3 command code can have 0xD3 or 0xD4 response code
- Error response frame will have 0xFF as the response code

Code	Instruction
0x03	Get Interrogator Module Information

0x22	Single Inventory
0x27	Multiple Inventory
0x28	Stop Multiple Inventory
0x0C	Set Select
0x12	Get Select
0x39	Read Tag
0x49	Write Tag
0x82	Lock Tag
0x65	Kill Tag
0x11	Set Communication Baud Rate
0x0D	Get Query Parameters
0x0E	Set Query Parameters
0x07	Set work area
0xAB	Set working channel
0xAA	Get working channel
0xAD	Set automatic frequency hopping
0xB7	Get transmit power
0xB6	Set transmit power
0xB0	Set to transmit continuous carrier
0xF1	Get receiver demodulator parameters
0xF0	Set the receiver demodulator parameters
0xF2	Test the blocking signal at the RF input
0xF3	Test channel RSSI
0x1A	Control IO port
0x17	Module sleep
0x1D	Set module idle sleep time
0xE0	NXP ChangeConfig command
0xE1	NXP ReadProtec/Reset ReadProtect command
0xE3	NXP Change EAS command
0xE4	NXP EAS-Alarm command
0xE5/0xE6	Impinj Monza 4 QT command
0xD3/0xD4	BlockPermalock instruction

Response Error Codes

If the execution of the instruction fails, the interrogator will return a response frame of execution failure which has 0xFF as the Command Code.

The response frame parameter will be 1 Byte of error code, and sometimes with PC+EPC of the tag if available.

Error	Code	Description
Command Error	0x17	The command code in the command frame is wrong.
FHSS Fail	0x20	Frequency hopping search channel timed out. All channels are occupied during this time.

Inventory Fail	0x15	No Tag reply or CRC error.
Wrong Access Password	0x16	Wrong access password.
Read Fail	0x09	No Tag reply or CRC error.
Write Fail	0x10	No Tag reply or CRC error.
Lock Fail	0x13	No Tag reply or CRC error.
Kill Fail	0x12	No Tag reply or CRC error.
BlockPermalock Fail	0x14	No Tag reply or CRC error.

NXP G2X ReadProtect Custom Command Error

Error	Code	Description
ChangeConfig Fail	0x1A	No Tag reply or CRC error.
ReadProtect Fail	0x2A	No Tag reply or CRC error.
Reset ReadProtect Fail	0x2B	No Tag reply or CRC error.
Change_EAS Fail	0x1B	No Tag reply or CRC error.
EAS_Alarm Fail	0x1D	The EAS_Alarm command fails, and the correct Alarm Code is returned without a tag.

EPC Gen2 Error

This is the format of the error response frame related to EPC Gen2 Protocol.

The error codes specified by the EPC Gen2 protocol is only 4 bits, and it is represented on the lower 4 bits of the response frame **error code**. The upper 4 bits is 0xA – 0xE depending on the command it corresponds to.

EPC Gen2 protocol returns an error code with this format:

Error Code Format	Command
0xA0 Error Code	Read
0xB0 Error Code	Write
0xC0 Error code	Lock
0xD0 Error code	Kill
0xE0 Error code	BlockPermalock

This is from Table I.2: Tag error codes of EPCglobal Gen2 Specification v2.0.1.

Error Code	Error-Code Name	Description
0x00	Other error	Catch-all for errors not covered by other codes.
0x01	Not supported	The Tag does not support the specified parameters or feature.
0x02	Insufficient privileges	The Interrogator did not authenticate itself with sufficient privileges for the Tag to perform the operation
0x03	Memory overrun	The Tag memory location does not exist, is too small, or the Tag does not support the specified EPC length.
0x04	Memory locked	The Tag memory location is locked or permalocked and is either not writeable or not readable.

0x05	Crypto suite error	Catch-all for errors specified by the cryptographic suite.
0x06	Command not encapsulated	The Interrogator did not encapsulate the command in an AuthComm or SecureComm as required.
0x07	ResponseBuffer overflow	The operation failed because the ResponseBuffer overflowed.
0x08	Security timeout	The command failed because the Tag is in a security timeout.
0x0B	Insufficient power	The Tag has insufficient power to perform the operation.
0x0F	Non-specific error	The Tag does not support error-specific codes.

Frame Examples

Interrogator Module Configuration

0x03 Get Interrogator module information

Get module information: hardware version, software version or manufacturer information.

Instruction Code: 0x03

Command Parameter:

- Hardware version: 0x00
- Software version: 0x01
- Manufacturer: 0x02

Response Parameter:

- First byte will be the same as the command parameter.
- Next bytes will be the hardware module name and version in ASCII encoding.

Command to get the hardware version

H	Type	CMD	PL	Parameter	CRC	End
BB	00	03	00 01	00	04	7E

Successful Response

H	Type	CM D	PL	Parameter	CR C	En d
B B	01	03	00 0B	00 4D 31 30 30 20 56 31 2E 30 30	22	7E

The response parameter starts with 0x00, corresponding to the command parameter. The next bytes are 4D 31 30 30 20 56 31 2E 30 30, which is ASCII code of "M100 V1.00"

0x11 Set communication baud rate

The baud rate in the parameter is the actual baud rate divided by 100. For this example, the target baud rate is 19200, so the value is $19200 / 100 = 192 = 0xC0$.

Command

H	Type	CMD	PL	Baud rate	CRC	End
BB	00	11	00 02	00 C0	D3	7E

No Response Frame

There is no response frame for this instruction. After the Interrogator has changed the communication baud rate, the Host Computer will need to reconnect to the Interrogator with the new baud rate.

0x17 Module Sleep

The module sleep command allows the Interrogator to maintain a low-power sleep mode. The Host Computer can wake the Interrogator up by sending any byte through the serial port, but the data will be discarded.

The first command received after the module sleeps will not respond because the first character of the first command will be abandoned. This command will reset the M100/QM100 chip after power-off. After the module wakes up, it will re-download the firmware to the M100/QM100 chip and reset some parameters to the module (these parameters include the power, frequency, and frequency hopping configured before sleep Mode, sleep time, receiver demodulator parameters, excluding Select mode, Select parameters, etc.), so some parameters may need to be reset. The instructions are as follows:

Command

H	Type	CMD	PL	CRC	End
BB	00	17	00 00	17	7E

Response

H	Type	CMD	PL	Parameter	CRC	End
BB	01	17	00 01	00	19	7E

0x1D Module Idle sleep time

This command can set how long the module will automatically enter the sleep state after no operation. After the module sleeps, it can wake up by sending any character through the serial port. After the module sleeps, the first command received will not respond because the first character of the first command will be discarded. This command will reset the M100/QM100 chip. After the module wakes up, it will re-download the firmware to the M100/QM100 chip and reset some parameters to the module (these parameters include the power, frequency, and frequency hopping mode configured before sleep. Sleep time, receiver demodulator parameters, excluding Select mode, Select parameters, etc.), so some parameters may need to be reset.

The parameter define number of minutes of inactivity before sleep. The range is 1 to 30 minutes. 0x00 means no sleep.

Command

To set the idle sleep time to two minutes:

H	Type	CMD	PL	Parameter	CRC	End
BB	00	1D	00 01	02	20	7E

Successful Response

H	Type	CMD	PL	Parameter	CRC	End
BB	01	1D	00 01	02	21	7E

0x04 Module Idle mode

This command allows the module to enter IDLE working mode. In IDLE mode, except for the digital part and communication interface, all other analog and RF parts of the power supply are turned off to reduce power consumption when not working. After the module enters the IDLE mode, it can still communicate with the module normally, the set parameters are still saved, and the module can normally respond to the instructions of the Host Computer. After entering IDLE mode, the first inventory (or instructions that require interaction with tags such as reading or writing tag data) will restore the module to its normal state, but the first inventory may be successful due to the unstable power state of the RF part. The rate drops, and subsequent inventory and other operations can return to normal.

First parameter: 0x00: Exit Idle Mode, 0x01: Enter Idle Mode

Second Parameter: Reserved as 0x01

Idle Mode Time: Number of minutes before entering Idle mode. The value ranges from 0x00 to 0x1E (30 Minutes). 0x00 means never enter Idle mode.

Command

To set the idle mode time to three minutes:

H	Type	CMD	PL	Enter	Reserved	Idle Time	CRC	End
B B	00	04	00 03	01	01	03	20	7E

Successful Response

H	Type	CMD	PL	Parameter	CRC	End
BB	01	04	00 01	00	06	7E

Inventory Commands**0x22 Single Inventory**

Complete an inventory operation in the EPC Class1 Gen2 protocol. The instruction does not include the Select operation. The power amplifier will be turned on and off automatically before and after each inventory command is executed. In the single inventory command, the Query operation parameter is configured by another command, and the initial value is already in the firmware.

Command

H	Type	CMD	PL	CRC	End
BB	00	22	00 00	22	7E

Notification

After the Interrogator receives a single inventory command, if it can read the tag with the correct CRC check, the Interrogator will return the data including RSSI, PC, EPC and CRC.

The Interrogator will return one notification frame for each of the successfully read Tag.

H	Type	CMD	PL	RSSI	PC	EPC	CRC	CRC	End
B B	02	22	00 11	C9	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	3A 76	EF	7E

The RSSI value reflects the size of the signal at the input of the chip, excluding antenna gain and directional coupler attenuation. RSSI is the signal strength at the input end of the chip, signed in hexadecimal, and the unit is dBm. In

the above example, the RSSI is 0xC9, which means the signal strength at the input of the chip is -55dBm. input end of the chip input end of the chip

0x15 Error Response

If no tag is detected or the data CRC error is detected, the parameter value will be error code 0x15:

H	Type	CMD	PL	Parameter	CRC	End
BB	01	FF	00 01	15	16	7E

0x27 Multiple Inventory

Command

Inventory multiple times, and the number of polling times is limited to 0-65535. If the number of polling is 10000 times, the command is as follows:

H	Type	CMD	PL	Reserved	CNT	CRC	End
BB	00	27	00 03	22	27 10	83	7E

0x2710 = 10,000.

Notification

The multiple inventory response frame is the same as the single inventory response frame:

H	Type	CMD	PL	RSSI	PC	EPC	CRC	CRC	End
BB	02	22	00 11	C9	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	3A 76	EF	7E

0x15 Error Response

If no tag is detected or the data CRC error is detected, the parameter value will be error code 0x15:

H	Type	CMD	PL	Parameter	CRC	End
BB	01	FF	00 01	15	16	7E

0x28 Stop multiple inventory

This instruction stops the multiple inventory in process.

Command

H	Type	CMD	PL	CRC	End
BB	00	28	00 00	28	7E

Success Response Frame

H	Type	CMD	PL	Parameter	CRC	End
BB	01	28	00 01	00	2A	7E

Select Commands

The parameters of this select commands are explained in 6.3.2.12.1 Select commands.

0x0C Set Select**Command:**

Set the Select parameter and set the Select mode to 0x02 at the same time (Send the Select command before inventory the label). In the case of multiple tags, you can only inventory, read and write operations for specific tags based on the Select parameter. E.g:

H	Type	CM D	PL	SelParam	Pointer	MaskLen	Truncate	Mask	CR C	End
BB	00	0C	00 13	01	00 00 00 20	60	00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	AD	7E

SelParam: 0x01 (Target: 0b000, Action: 0b000, MemBank: 0b01)

SelParam has a total of 1 Byte, of which Target occupies the highest 3 bits, Action occupies the middle 3 bits, and MemBank occupies the last 2 bits.

- **Target**
Refer to EPC Gen2 Protocol
- **Action**
Refer to EPC Gen2 Protocol
- **Membank**
The meaning of MemBank is as follows:
 - 0b00: Reserved Memory Bank
 - 0b01: EPC Memory Bank
 - 0b10: TID Memory Bank
 - 0b11: User Memory Bank

Pointer: 0x00000020 (in bits, not words) Starting from the EPC Memory Bank

MaskLen: 0x60 (6 words, 96 bits))

Whether Truncate: 0x00 (0x00 is Disable truncation, 0x80 is Enable truncation)

When the Select Mask length is greater than 80 bits (5 words), sending the Select command will first set all tags in the field to Inventoried Flag as A and SL Flag as ~SL, and then operate according to the selected Action. When the Select Mask length is less than 80 bits (5 words), the label status will not be set to Inventoried Flag A and SL Flag ~SL through the Select command in advance.

Mask: 0x30751FEB705C5904E3D50D70

Success Response:

H	Type	CMD	PL	Data	CRC	End
BB	01	0C	00 01	00	0E	7E

0x0B Get Select**Command:**

H	Type	CMD	PL	CRC	End
BB	00	0B	00 00	0B	7E

Response:

H	Type	CMD	PL	Se IParam	P ointer	M askLen	Tr uncate	Mask	CRC	End
BB	01	0B	00 13	01	00 00 00 20	60	00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	AD	7E

SelfParam: 0x01 (Target: 3'b000, Action: 3'b000, MemBank: 2'b01)

Pointer: 0x00000020 (in bits, not words) Starting from the EPC Memory Bank

MaskLen: 0x60 (6 words, 96 bits))

Whether Truncate: 0x00 (0x00 is Disable truncation, 0x80 is Enable truncation)

0x12 Set Select mode**Command:**

If the Select parameter has been set, execute this command to set the Select mode. For example, if you want to cancel the Select command:

H	Type	CMD	PL	Mode	CRC	End
BB	00	12	00 01	01	14	7E

Select Mode Mode:

0x00: Send the Select command in advance to select a specific label before all operations on the label.

0x01: Do not send the Select command before operating on the label.

0x02: Only send the Select command before the tag operation except Inventory, such as
Before Read, Write, Lock, Kill, select a specific label through Select.

Success Response:

H	Type	CMD	PL	Data	CRC	End
BB	01	0C	00 01	00	0E	7E

RFID Tag Commands**0x39 Read Tag memory**

Read the data of the specified address and length in the memory Bank of a tag. This instruction takes four parameters:

- Access Password
- MemBank
- WordPtr
- WordCount

Before this instruction, the Select parameter should be set to select the specified tag to read the tag data area. If the Access Password is all zeros, no Access command is sent.

Command

This command reads from 2 words (4 bytes) starting from address 0x00 of User Memory Bank using 0x0000FFFF as Access Password.

H	Type	CMD	PL	Access Password	MemBank	WordPtr	WordCount	CRC	End
BB	00	39	00 09	00 00 FF FF	03	00 00	00 02	45	7E

Successful Response

H	Type	CMD	PL	PC + EPC Length	PC	EPC	Read Data	CRC	End
BB	01	39	00 13	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	12 34 56 78	B0	7E

0x09 Read Fail Response

H	Type	CMD	PL	Error Code	CRC	End
BB	01	FF	00 01	09	0A	7E

0x09 error code means the tag is not in range, or there is an error of the EPC code.

0x16 Wrong Access Password Error

H	Type	CMD	PL	Error Code	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	16	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	75	7E

0xA3 EPC Gen2 Error Response:

H	Type	CMD	PL	Error Code	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	A3	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	02	7E

Error Code 0xA3 means that this is an EPC Gen2 error code 0x03: Memory Overrun.

0x49 Write Tag memory

Before this instruction, the Select parameter should be set first to select the specific tag to write to. If the Access Password is all zeros, no Access command is sent.

The WordCount should not exceed 32 words (64 Bytes = 512 bits).

Command

H	Type	CMD	PL	Access Password	MemBank	WordPtr	WordCount	Data	CRC	End
BB	00	49	00 0D	00 00 FF FF	03	00 00	00 02	12 34 56 78	6D	7E

Successful Response

H	Type	CMD	PL	PC + EPC Length	PC	EPC	Parameter	CRC	End
BB	01	49	00 10	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	00	A9	7E

Parameter 0x00 means that the instruction is executed successfully.

0x10 Error Response

H	Type	CMD	PL	Error	CRC	End
BB	01	FF	00 01	10	0A	7E

0x10 error code means the tag is not in range, or there is an error of the EPC code.

0x16 Wrong Access Password Error

H	Type	CMD	PL	Error	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	16	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	75	7E

0xB3 EPC Gen2 Error Response

H	Type	CMD	PL	Error	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	B3	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	12	7E

Error code 0xB3 means that this is an EPC Gen2 error code 0x03: Memory Overrun.

0x82 Lock Tag Memory

Change lock status of a memory bank of the specified tag. Before this instruction, the Select parameter should be set to target the specified tag.

Command

H	Type	CMD	PL	Access Password	LD	CRC	End
B B	00	82	00 07	00 00 FF FF	02 00 80	09	7E

In this command, the LD is 0x020008.

0x020080 = 0b0000 0b0010 0b0000 0b0000 0b1000 0b0000

The upper 4 bits of the LD is reserved to 0b0000 because the EPC Gen2 Lock-Command Payload only requires 20 bits of data. (Refer to section 6.3.2.11.3.5 of EPC Gen2 Protocol v2.0.1)

Only actions whose mask bits are 1 are going to be executed.

In this 0x020080 example, the mask bits of Access Password is 01 and the action of Access Password is 01. Thus, the access password will be permanently readable and writable.

Successful Response

H	Type	CMD	PL	PC + EPC Length	PC	EPC	Parameter	CRC	End
BB	01	82	00 10	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	00	E2	7E

0x13 Error Response

H	Type	CMD	PL	Parameter			CRC	End
BB	01	FF	00 01	13			14	7E

0x13 error code means the tag is not in range, or there is an error of the EPC code.

0x16 Wrong Access Password Error

H	Type	CMD	PL	Error Code	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	16	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	75	7E

0xC4 EPC Gen2 Error Response

H	Type	CMD	PL	Error Code	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	C4	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	23	7E

Error Code 0xC4 means that this is an EPC Gen2 error code 0x04: Memory Locked.

0x65 Kill Tag

Before this instruction, the Select parameter should be set first to select the specific tag to kill.

Command

H	Type	CMD	PL	Kill Password			CRC	End
BB	00	65	00 04	00 00 FF FF			67	7E

Successful Response

H	Type	CMD	PL	PC + EPC Length	PC	EPC	Parameter	CRC	End
BB	01	65	00 10	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	00	C5	7E

0x12 Error Response

H	Type	CMD	PL	Parameter			CRC	End
BB	01	FF	00 01	12			13	7E

0x12 error code means the tag is not in range, or there is an error of the EPC code.

0xD0 EPC Gen2 Error Response

H	Type	CMD	PL	Error Code	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	D0	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	2F	7E

Error Code 0xD0 means that this is an EPC Gen2 error code 0x00: Other error. This may be because the kill password of the tag hasn't been set.

0xD3 BlockPermalock / 0xD4 Read Permalock Status

This instruction can permanently lock certain blocks in the user area, or read the lock status of the block. Before this instruction, the Select parameter should be set to the targeted tag.

According to section 6.3.2.12.3.9 of EPCglobal Gen2 Specification v2.0.1:

A BlockPermalock may permalock between zero and 4080 memory blocks. The block size, which is predefined by the Tag manufacturer, is fixed at between one and 1024 words, is the same for all files, and is the same for block permalocking and file allocation. The memory blocks specified by a BlockPermalock need not be contiguous.

A Tag shall only execute a BlockPermalock in the secured state.

A BlockPermalock differs from a Lock in that BlockPermalock permanently locks individual blocks of FileN, $N > 0$ of User memory in an unwriteable state whereas Lock reversibly or permanently locks the kill and/or access password, the EPC memory bank, the TID memory bank, and/or File0 of User memory in a writeable or un-writeable state. Table 6.55 specifies how a Tag shall behave upon receiving a BlockPermalock targeting File_0 that follows a prior Lock, or vice versa (assuming Read/Lock=1).

Parameter	Length	Explanation
Access Password	4 Bytes	Access password of the tag
Read / Lock	1 Byte	0x00: Read 0x01: Write
MemBank	1 Byte	Should be 0x03 (User memory)
BlockPtr	2 Byte	BlockPtr specifies the starting address for Mask, in units of 16 blocks
BlockRange	1 Byte	BlockRange specifies the range of Mask, starting at BlockPtr and ending (16×BlockRange)–1 blocks later.
Mask	0 / 2 Byte	If the instruction is to read (0x00 for Read/Lock Parameter), this parameter is omitted

Command

To permanently lock the blocks 5, 6, 7:

H	Type	CMD	PL	Access Password	Read / Lock	MemBank	BlockPtr	BlockRange	Mask	CRC	End
BB	00	D3	00 0B	00 00 FF FF	01	03	00 00	01	07 00	E8	7E

0xD3 Read Successful Response

H	Type	CMD	PL	PC + EPC Length	PC	EPC	BlockRange	Block Permalock	CRC	End
---	------	-----	----	-----------------	----	-----	------------	-----------------	-----	-----

BB	01	D3	00 12	0E	30 00	E2 00 30 16 66 06 00 69 11 60 9F 94	01	07 00	CD	7E
----	----	----	-------	----	-------	--	----	-------	----	----

0xD4 Write Successful Response

H	Type	CMD	PL	PC + EPC Length	PC	EPC	Parameter	CRC	End
BB	01	D4	00 10	0E	30 00	E2 00 30 16 66 06 00 69 11 60 9F 94	00	C4	7E

0x14 Tag Not Found Error

H	Type	CMD	PL	Parameter	CRC	End
BB	01	FF	00 01	14	15	7E

0xE3 EPC Gen2 Error Response

H	Type	CMD	PL	Error	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	E3	0E	30 00	E2 00 30 16 66 06 00 69 11 60 9F 94	D2	7E

Error Code 0xE3 means that this is an EPC Gen2 error code 0x03: Memory Overrun.

0x16 Wrong Access Password Error

H	Type	CMD	PL	Error Code	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	16	0E	30 00	E2 00 30 16 66 06 00 69 11 60 9F 94	05	7E

Query Commands

Query command is documented on section 6.3.2.12.2 of EPC Gen2 Protocol v2.0.1.

The parameters for communication between the Host Computer and the Interrogator is documented in the table below with the data represented in binary:

Description	Length (bit)	Description
DR	1	0 = 8
M	2	00 = 1, 01 = 2, 10 = 4, 11 = 8
TRext	1	0 = No pilot tone, 1 = Use pilot tone
Sel	2	00 = All, 01 = All, 10 = ~SL, 11 = SL
Session	2	00 = S0, 01 = S1, 10 = S2, 11 = S3
Target	1	0 = A, 1 = B
Q (slot-count)	4	Decimal 0-15 according to the value
Padding	3	000

0x0D Get Query parameters**Command**

H	Type	CMD	PL	CRC	End
BB	00	0D	00 00	0D	7E

Successful Response

H	Type	CMD	PL	Parameter	CRC	End
BB	01	0D	00 02	10 20	40	7E

The parameter is 2 bytes, and the following specific parameters are spliced bit by bit. The response parameter corresponding to the above response frame is 0x1020 = 0b0001000000100000

Refer to section 6.3.2.12.2.1 EPC Gen2 Protocol v2.0.1

In the table below, the data are reresented in binary.

Description	Length (bit)	Data	Description
DR	1	0	0 = 8
M	2	00	00 = 1
TRExt	1	1	1 = Use pilot tone
Sel	2	00	00 = All
Session	2	00	00 = S0
Target	1	0	0 = A
Q (slot-count)	4	0100	4
Padding	3	000	

DR=8, M=1, TRExt=Use pilot tone, Sel=All, Session=S0, Target=A, Q=4

0x0E Set Query parameters**Command**

H	Type	CMD	PL	Parameter	CRC	End
BB	00	0E	00 02	10 20	40	7E

This command set the Query parameters to the same value as the previous get Query parameters instruction example.

Successful Response

H	Type	CMD	PL	Parameter	CRC	End
BB	01	0E	00 01	00	10	7E

Radio Frequency

Radio frequency of the Interrogator is determined by the region and the frequency channel listed below.

Frequency Region	Region Index	Frequency Channel in MHz	CH_INDEX
------------------	--------------	--------------------------	----------

920++ MHz Based	01	$(CH_Index * 0.25 + 920.125) \text{ MHz}$	$(Freq_CH - 920.125) / 0.25$
United States	02	$(CH_Index * 0.5 + 902.25) \text{ MHz}$	$(Freq_CH - 902.25) / 0.5$
Europe	03	$(CH_Index * 0.2 + 865.1) \text{ MHz}$	$(Freq_CH - 865.1) / 0.2$
840++ MHz Based	04	$(CH_Index * 0.25 + 840.125) \text{ MHz}$	$(Freq_CH - 840.125) / 0.25$
Korea	06	$(CH_Index * 0.2 + 917.1) \text{ MHz}$	$(Freq_CH - 917.1) / 0.2$

0x07 Set Frequency Region

To set the Frequency Region 920++ MHz Based, set the Region Index to 0x01.

Command

H	Type	CMD	PL	Region Index	CRC	End
BB	00	07	00 01	01	09	7E

Successful Response

H	Type	CMD	PL	Parameter	CRC	End
BB	01	07	00 01	00	09	7E

0x08 Get Frequency Region

Command

H	Type	CMD	PL	CRC	End
BB	00	08	00 00	08	7E

Successful Response

H	Type	CMD	PL	Region Index	CRC	End
BB	01	08	00 01	01	09	7E

The region index is 01, so the frequency region is 920++ MHz Based.

0xAB Set Frequency Channel

If it is the 900++ MHz Based, to set the frequency channel of the reader to 920.375 MHz, set the CH_Index to: $(920.375 - 920.125) / 0.25 = 01$.

Command

H	Type	CMD	PL	CH_Index	CRC	End
BB	00	AB	00 01	01	AD	7E

Successful Response

H	Type	CMD	PL	Parameter	CRC	End
BB	01	AB	00 01	00	AD	7E

0xAA Get Frequency Channel**Command**

H	Type	CMD	PL	CRC	End
BB	00	AA	00 00	AA	7E

Successful Response

H	Type	CMD	PL	CH_Index	CRC	End
BB	01	AA	00 01	00	AC	7E

If the frequency region is 920++ MHz based, the CH_Index 0x00 means the frequency channel is: $(0 * 0.25 + 920.125)$ MHz = 920.125 MHz.

0xAD Set automatic frequency hopping

In the automatic frequency hopping mode, if the user executes the instruction to insert the working channel, the reader will randomly select the channel frequency hopping from the channel list set by the user, otherwise, the channel frequency hopping will be randomly selected according to the internal preset channel list.

Possible parameter for the command is:

- 0x00: Disable automatic frequency hopping
- 0xFF: Enable automatic frequency hopping

Command

H	Type	CMD	PL	Parameter	CRC	End
BB	00	AD	00 01	FF	AD	7E

Successful Response

H	Type	CMD	PL	Parameter	CRC	End
BB	01	AD	00 01	00	AF	7E

0xA9 Insert working channel

Inserting the working channel allows the user to independently set the frequency hopping channel list. After executing this command, the reader will randomly select the channel hopping frequency from the channel list set by the user. The command is defined as follows:

Command

H	Type	CMD	PL	CH Count	CH List	CRC	End
B B	00	A9	00 06	05	01 02 03 04 05	C3	7E

Successful Response

H	Type	CMD	PL	Parameter	CRC	End
BB	01	A9	00 01	00	AB	7E

0xB7 Get transmit power**Command**

H	Type	CMD	PL	CRC	End
BB	00	B7	00 00	B7	7E

Response

H	Type	CMD	PL	Power	CRC	End
BB	01	B7	00 02	07 D0	91	7E

Power: 0x07D0 = Decimal 2000 = 20 dBm

0xB6 Set Transmit Power**Command**

H	Type	CMD	PL	Power	CRC	End
BB	00	B6	00 02	07 D0	8F	7E

Successful Response

H	Type	CMD	PL	Parameter	CRC	End
BB	01	B6	00 01	00	B8	7E

0xB0 Set to transmit continuous wave

Parameter:

- 0x00: Disable Continuous Wave
- 0xFF: Enable Continuous Wave

Command

H	Type	CMD	PL	Parameter	CRC	End
BB	00	B0	00 01	FF	B0	7E

Successful Response

H	Type	CMD	PL	Parameter	CRC	End
BB	01	B0	00 01	00	B2	7E

0xF1 Get receiver demodulator parameters

Includes: Mixer gain, IF AMP gain and signal demodulation threshold.

Command

H	Type	CMD	PL	CRC	End
BB	00	F1	00 00	F1	7E

Response

H	Type	CMD	PL	Mixer_G	IF_G	Threshold	CRC	End
BB	01	F1	00 04	03	06	01 B0	B0	7E

Mixer gain Mixer_G: 0x03 (Mixer gain is 9dB)

IF amplifier gain IF_G: 0x06 (IF AMP gain is 36dB)

Signal demodulation threshold Thrd: 0x01B0 (the smaller the signal demodulation threshold, the lower the return RSSI of the demodulated tag, but the more unstable it is. It cannot be demodulated at all if the value is lower than a certain value; on the contrary, the larger the threshold, the demodulated tag return signal RSSI The larger the distance, the more stable. 0x01B0 is the recommended minimum)

0xF0 Set the receiver demodulator parameters

Mixer Gain Index	Mixer Gain Value (dB)
0x00	0
0x01	3
0x02	6
0x03	9
0x04	12
0x05	15
0x06	16

IF AMP Gain Index	IF AMP Gain Value (dB)
0x00	12
0x01	18
0x02	21
0x03	24
0x04	27
0x05	30
0x06	36
0x07	40

Command

H	Type	CMD	PL	Mixer Gain Index	IF AMP Gain Index	Threshold	CRC	End
BB	00	F0	00 04	03	06	01 B0	AE	7E

Successful Response

H	Type	CMD	PLParameter	CRC	End
---	------	-----	-------------	-----	-----

BB	01	F0	00 01	F2	7E
----	----	----	-------	----	----

0xF2 Test the blocking signal at the RF input

Testing the blocking signal at the RF input Scan Jammer is used to detect the magnitude of the blocking signal of each channel of the reader antenna in the current area.

Command

H	Type	CMD	PL	CRC	End
BB	00	F2	00 00	F2	7E

Response

H	Type	CMD	PL	CH_L	CH_H	JMR	CRC	End
BB	01	F2	00 16	00	13	F2 F1 F0 EF EC EA E8 EA EC EE F0 F1 F5 F5 F5 F6 F5 F5 F5 F5	EA	7E

If there are a total of 20 channels in the 900MHz frequency band in China, and the scanning of the radio frequency input blocking signal Scan Jammer channel is executed correctly

Test start channel CH_L: 0x00 (Test start channel Index is 0)

Test end channel CH_H: 0x13 (Test end channel Index is 19)

Channel blocking signal JMR: 0xF2F1F0EFECEAE8EAECEEF0F1F5F5F5F6F5F5F5F5 (blocking signal for each channel)

JMR is represented by a signed Byte, where 0xF2 is -14dBm).

0xF3 Test channel RSSI

Command

H	Type	CMD	PL	CRC	End
BB	00	F3	00 00	F3	7E

Response

H	Type	CMD	PL	CH_L	CH_H	RSSI	CRC	End
BB	01	F3	00 16	00	13	BA BA BA BA BA BA BA BA BA BA BA BA BA BA BA BA BA BA BA BA	A5	7E

If there are 20 channels in 900MHz Based Frequency, this is the returned RSSI for each channel.

Each channel RSSI is represented by a signed Byte, where 0xBA is -70 dBm.

NXP Custom Commands

0xE1 NXP ReadProtect/Reset ReadProtect Command

NXP G2X tags support ReadProtect/Reset ReadProtect commands. When the tag executes the ReadProtect command successfully, the ProtectEPC and ProtectTID bits of the tag will be set to '1' and the tag will enter the data protection state. If the label returns to the normal state from the data protection state, the Reset ReadProtect command needs to be executed. Before this instruction, the Select parameter should be set to select the specified label for operation.

Parameter	Length	Explanation
Access Password	4 Bytes	Access Password of the RFID Tag
Set / Reset	1 Byte	0x00: Set Read Protect 0x01: Reset Read Protect

Command

To set read protect to the selected tag with 0x0000FFFF access password:

H	Type	CM D	PL	Access Password	Set / Reset	CR C	End
B B	00	E1	00 05	00 00 FF FF	00	E4	7E

Successful Response

H	Type	CMD	PL	PC + EPC Length	PC	EPC	Pa rameter	CRC	End
BB	01	E1	00 10	0E	30 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	00	3D	7E

Successful Reset Read Protect Response

H	Type	CMD	PL	PC + EPC Length	PC	EPC	Pa rameter	CRC	End
BB	01	E2	00 10	0E	30 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	00	3E	7E

0x2A Tag Not Found when Setting Read Protect

H	Type	CMD	PL	Parameter	CRC	End
BB	01	FF	00 01	2A	2B	7E

0x2A error code means the tag is not in range, or there is an error of the EPC code.

0x2B Tag not Detected when Resetting Read Protect

H	Type	CMD	PL	Parameter	CRC	End
BB	01	FF	00 01	2B	2C	7E

0x16 Wrong Access Password Error

H	Type	CMD	PL	Error Code	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	16	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	75	7E

0xE3 NXP Change EAS

NXP G2X tags support the Change EAS command. When the tag executes the Change EAS command successfully, the PSF bit of the tag will change to '1' or '0' accordingly. When the PSF position of the tag is '1', the tag will respond to the *EASAlarm instruction*, otherwise the tag will not respond to the *EASAlarm instruction*. Before this instruction, the Select parameter should be set to select the specified label for operation.

Command

To set PSF bit to 1 (Tag will response to EAS_Alarm):

H	Type	CMD	PL	Access Password	PSF	CRC	End
BB	00	E3	00 05	00 00 FF FF	01	E7	7E

Successful Response

H	Type	CMD	PL	PC + EPC Length	PC	EPC	Parameter	CRC	End
BB	01	E3	00 10	0E	30 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	00	3F	7E

0x1B Tag Not Found Error

H	Type	CMD	PL	Parameter	CRC	End
BB	01	FF	00 01	1B	1C	7E

0x16 Wrong Access Password Error

H	Type	CMD	PL	Error Code	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	16	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	75	7E

0xE4 NXP EAS_Alarm

Tags with PSF bit set to 1 will respond to this instruction by returning 64-bit EAS Alarm code.

Command

H	Type	CMD	PL	CRC	End
BB	00	E4	00 00	E4	7E

Successful Response

H	Type	CMD	PL	EAS Alarm Code	CRC	End
---	------	-----	----	----------------	-----	-----

B B	01	E4	00 08	69 0A EC 7C D2 15 D8 F9	80	7E
--------	----	----	-------	-------------------------	----	----

0x1D No Tag Responding

H	Type	CMD	PL	Parameter	CRC	End
BB	01	FF	00 01	1D	1E	7E

0xE0 NXP ChangeConfig

Some series of NXP G2X tags (such as G2iM and G2iM+) support the ChangeConfig command, which can be used to read or modify the 16bits Config-Word of NXP G2X tags. The Config-Word of the NXP G2X tag is located at the address 20h (word address) of EPC Memory Bank and can be read by a normal Read command. When the tag is in the Secured state (safe state), the Config-Word of the tag can be rewritten. It should be noted that rewriting Config-Word means flipping the corresponding data bit of Config-Word, that is, writing the corresponding bit of '1' flipping ('1' 'Becomes '0', '0' becomes '1'), the corresponding bit written in '0' remains unchanged. Before this instruction, the Select parameter should be set to select the specified label for operation.

Config parameter: when 0x0000 means read.

Command

H	Type	CMD	PL	Access Password	Config	CRC	End
B B	00	E0	00 06	00 00 FF FF	00 00	E4	7E

Successful Response

H	Type	CMD	PL	PC + EPC Length	PC	EPC	Config	CRC	End
BB	01	E0	00 11	0E	30 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	00 41	7E	7E

0x1A No Tag Responding

H	Type	CMD	PL	Parameter	CRC	End
BB	01	FF	00 01	1A	1B	7E

0x16 Wrong Access Password Error

H	Type	CMD	PL	Error Code	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	16	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	75	7E

Impinj Monza Custom Commands

0xE5 Impinj Monza QT Command

Impinj Monza 4 QT tags support the QT command, which can modify the QT Control word of the tag. Setting the QTSR bit can shorten the operation distance of the tag in the Open and Secured states or when it is about to enter the Open and Secured states. Modifying the QTMEM bit can switch the label to use Public Memory Map (Public Memory Map) or Private Memory Map (Private Memory Map). Before this instruction, the Select parameter should be set to select the specified label for operation.

Parameter	Length	Explanation
Access Password	4 Bytes	Access password of the tag
Read / Write	1 Byte	0x00: Read 0x01: Write
Persistence	1 Byte	0x00: Write to volatile memory 0x01: Write to NVM memory
Payload	2 Byte	QT Control: QTSR and QTMEM

Command

To write to NVM memory with QTSR 0x40 and QTMEM 0x00:

H	Type	CM D	PL	Access Password	Read / Write	Persistence	Payload	CR C	End
B B	00	E5	00 08	00 00 FF FF	01	01	40 00	2D	7E

0xE5 Read Successful Response

H	Type	CMD	PL	PC + EPC Length	PC	EPC	QT Control	CRC	End
BB	01	E5	00 11	0E	30 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	00 00	42	7E

0xE6 Write Successful Response

H	Type	CMD	PL	PC + EPC Length	PC	EPC	Pa rameter	CRC	End
BB	01	E6	00 10	0E	30 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	00	42	7E

0x2E Tag Not Found Error

H	Type	CMD	PL	Parameter	CRC	End
BB	01	FF	00 01	2E	2F	7E

0x16 Wrong Access Password Error

H	Type	CMD	PL	Error Code	PC + EPC Length	PC	EPC	CRC	End
BB	01	FF	00 10	16	0E	34 00	30 75 1F EB 70 5C 59 04 E3 D5 0D 70	75	7E